Eqoustics Manual

Reference instructions for using Eqoustics.

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Eqoustics is a notebook-style mathematics editor. Each row is stored as LaTeX in the background and displayed as readable notation in the editor.

Documents

- Use `File > New Document` to start a blank notebook.
- Use `File > Open Existing Document` to open a saved `.tex` notebook.
- Use `File > Save File` to save the current notebook.
- Use `File > Save As` to save a copy or choose a new file location.
- Use `File > Export As > PDF` to export a printable PDF.
- Use `File > Export As > LaTeX` to save the notebook as a LaTeX `.tex` file.
- Recent files appear in the File menu. Select one to reopen it quickly.

Editing

- Type directly into the active row.
- Use the row controls to add, move, or remove rows.
- Use `Edit > Undo` and `Edit > Redo` to move backward and forward through document changes.
- Use `Edit > Cut`, `Copy`, and `Paste` for normal clipboard work.
- Use `Edit > Copy As` to copy selected rows as LaTeX, notation, PNG, or JPEG.

Mathematical notation

- The notation toolbar inserts common math structures and symbols.
- Use fractions for expressions like `\frac{x+1}{2}`.
- Use powers for exponents like `x^2` and subscripts like `a_n`.
- Use roots for square roots and indexed roots, such as `\sqrt{x}` and `\sqrt[3]{x}`.
- Use relation symbols such as `=`, `<`, `>`, `\leq`, and `\geq`.
- Use Greek letters such as `\alpha`, `\beta`, `\theta`, and `\pi`.
- Use matrices and cases from the toolbar when expressions need rows and columns.

Formatting

- Use the bold, italic, and underline buttons for inline text formatting.
- Use heading buttons `H1` through `H4` to structure a notebook.
- Use bullet and numbered list buttons for list formatting.
- Use the hyperlink button to add a link to selected text.

Microphone

The floating microphone window controls speech recognition. Click the microphone to start listening. Eqoustics records speech chunks, sends them to Gemma, then inserts the result into the active row.

When speech starts for the first time, Gemma may need to load. If the selected runtime uses Transformers, the Gemma model may also need to download from Hugging Face into the local cache. Loading or downloading can take time, especially on first run. The speech control shows progress while the model is preparing.

Speech commands

Speech commands control the notebook without typing. Open the `Commands` menu to see the live command list and aliases. Common commands include:

- `new line`, `newline`, `new row`, or `next line` to create a row below the current row.
- `new paragraph` or `next paragraph` to create a blank row for a new paragraph.
- `go to row five`, `go to line five`, `select row five`, or `select line five` to move to a numbered row.
- `silence` to keep the microphone active while ignoring dictation.
- `listen` to resume processing after silence mode.
- `deactivate microphone`, `disable microphone`, `turn off microphone`, or `stop listening` to turn speech recognition off.

Dictating LaTeX

Speak mathematical expressions naturally. Eqoustics asks Gemma to convert math dictation to LaTeX, then inserts the rendered notation.

- Say `x squared over two minus three` to insert a fraction and exponent.
- Say `x equals y` to insert an equation.
- Say `plus five` to append an operator expression.
- Say `make that a plus instead of minus` to correct the current row when the command depends on existing row content.

Dictating text

If speech is plain non-math prose, Eqoustics inserts it as text instead of LaTeX. For example, saying `that is the conclusion of the proof` inserts normal text into the active row.

If Gemma returns math, Eqoustics inserts LaTeX notation. If Gemma returns text, Eqoustics inserts text. If the result is a command, Eqoustics runs the command instead of inserting content.

Mathematical notation reference

This table is generated from the toolbar source by `scripts/update-help-notation-table.mjs`. Run it after changing toolbar notation definitions.

Notation as shown	Called	LaTeX inserted	Menu
□/□	Fraction	`\frac{□}{□}`	Quick notation
√□	Square root	`\sqrt`	Quick notation
^{n}√□	Indexed root	`\sqrt[]`	Quick notation
x^□	Power / exponent	`^`	Quick notation
x_□	Subscript	`_`	Quick notation
□ under x	Lower bound	`\underset`	Quick notation
□ over x	Upper bound	`\overset`	Quick notation
□ over and under x	Lower and upper bounds	`\bothset`	Quick notation
|□|	Absolute value	`\left\lvert{□}\right\rvert`	Quick notation
(□)	Parentheses	`\left({□}\right)`	Quick notation
[□]	Brackets	`\left[{□}\right]`	Quick notation
{□}	Curly braces	`\left\lbrace{□}\right\rbrace`	Quick notation
+	Plus	`+`	Arithmetic
-	Minus	`-`	Arithmetic
×	Multiplication (cross)	`\times`	Arithmetic
·	Multiplication (dot)	`\cdot`	Arithmetic
÷	Division	`\div`	Arithmetic
∂	Partial derivative	`\partial`	Calculus
∇	Nabla / Del	`\nabla`	Calculus
∞	Infinity	`\infty`	Calculus
dy/dx	Derivative	`\frac{dy}{dx}`	Calculus
\int_{ }^{ }	Definite integral — type the bounds	`\int_{ }^{ }`	Calculus
Custom	Plain matrix with custom delimiters	`\begin{matrix} □ & □ \\ □ & □ \end{matrix}`	Matrix environments
matrix	Plain matrix with no brackets	`\begin{matrix} □ & □ \\ □ & □ \end{matrix}`	Matrix environments
pmatrix	Matrix with parentheses	`\begin{pmatrix} □ & □ \\ □ & □ \end{pmatrix}`	Matrix environments
bmatrix	Matrix with square brackets	`\begin{bmatrix} □ & □ \\ □ & □ \end{bmatrix}`	Matrix environments
vmatrix	Matrix with single vertical bars	`\begin{vmatrix} □ & □ \\ □ & □ \end{vmatrix}`	Matrix environments

Notation as shown	Called	LaTeX inserted	Menu
Vmatrix	Matrix with double vertical bars	`\begin{Vmatrix} □ & □ \\ □ & □ \end{Vmatrix}`	Matrix environments
Bmatrix	Matrix with curly braces	`\begin{Bmatrix} □ & □ \\ □ & □ \end{Bmatrix}`	Matrix environments
smallmatrix	Compact inline matrix	`\begin{smallmatrix} □ & □ \\ □ & □ \end{smallmatrix}`	Matrix environments
array	Column-aligned array with column spec	`\begin{array}{ccc} □ & □ \\ □ & □ \end{array}`	Matrix environments
dotted	Array with dotted cell dividers	`\begin{array}{ccc} □ & □ \\ □ & □ \end{array}`	Matrix environments
None	Matrix delimiter: None	`no delimiter`	Matrix delimiters
Parentheses	Matrix delimiter: Parentheses	`\left( ... \right)`	Matrix delimiters
Brackets	Matrix delimiter: Brackets	`\left[ ... \right]`	Matrix delimiters
Braces	Matrix delimiter: Braces	`\left\lbrace ... \right\rbrace`	Matrix delimiters
Angles	Matrix delimiter: Angles	`\left\langle ... \right\rangle`	Matrix delimiters
Vertical bars	Matrix delimiter: Vertical bars	`\left\lvert ... \right\rvert`	Matrix delimiters
Double bars	Matrix delimiter: Double bars	`\left\lVert ... \right\rVert`	Matrix delimiters
Ceiling	Matrix delimiter: Ceiling	`\left\lceil ... \right\rceil`	Matrix delimiters
Floor	Matrix delimiter: Floor	`\left\lfloor ... \right\rfloor`	Matrix delimiters
Groups	Matrix delimiter: Groups	`\left\lgroup ... \right\rgroup`	Matrix delimiters
Moustaches	Matrix delimiter: Moustaches	`\left\lmoustache ... \right\rmoustache`	Matrix delimiters
Upper corners	Matrix delimiter: Upper corners	`\left\ulcorner ... \right\urcorner`	Matrix delimiters
Lower corners	Matrix delimiter: Lower corners	`\left\llcorner ... \right\lrcorner`	Matrix delimiters
Double brackets	Matrix delimiter: Double brackets	`\left\llbracket ... \right\rrbracket`	Matrix delimiters
Up arrows	Matrix delimiter: Up arrows	`\left\uparrow ... \right\uparrow`	Matrix delimiters
Down arrows	Matrix delimiter: Down arrows	`\left\downarrow ... \right\downarrow`	Matrix delimiters
Up-down arrows	Matrix delimiter: Up-down arrows	`\left\updownarrow ... \right\updownarrow`	Matrix delimiters
Double up arrows	Matrix delimiter: Double up arrows	`\left\Uparrow ... \right\Uparrow`	Matrix delimiters

Notation as shown	Called	LaTeX inserted	Menu
Double down arrows	Matrix delimiter: Double down arrows	`\left\Downarrow ... \right\Downarrow`	Matrix delimiters
Double up-down arrows	Matrix delimiter: Double up-down arrows	`\left\Updownarrow ... \right\Updownarrow`	Matrix delimiters
Auto	Matrix delimiter size: Auto	`\left ... \right`	Matrix delimiter sizes
Small	Matrix delimiter size: Small	`\bigl ... \bigr`	Matrix delimiter sizes
Medium	Matrix delimiter size: Medium	`\Bigl ... \Bigr`	Matrix delimiter sizes
Large	Matrix delimiter size: Large	`\biggl ... \biggr`	Matrix delimiter sizes
Largest	Matrix delimiter size: Largest	`\Biggl ... \Biggr`	Matrix delimiter sizes
cases	Left-braced cases	`\begin{cases} □ & \text{if } □ \\ □ & \text{if } □ \end{cases}`	Cases
rcases	Right-braced cases	`\begin{rcases} □ & \text{if } □ \\ □ & \text{if } □ \end{rcases}`	Cases
α	Alpha	`\alpha`	Greek
β	Beta	`\beta`	Greek
γ	Gamma	`\gamma`	Greek
δ	Delta	`\delta`	Greek
ε	Epsilon	`\epsilon`	Greek
ϵ	Variant epsilon	`\varepsilon`	Greek
ζ	Zeta	`\zeta`	Greek
η	Eta	`\eta`	Greek
θ	Theta	`\theta`	Greek
ϑ	Variant theta	`\vartheta`	Greek
ι	Iota	`\iota`	Greek
κ	Kappa	`\kappa`	Greek
ϰ	Variant kappa	`\varkappa`	Greek
λ	Lambda	`\lambda`	Greek
μ	Mu	`\mu`	Greek
ν	Nu	`\nu`	Greek
ο	Omicron	`o`	Greek
ξ	Xi	`\xi`	Greek
π	Pi	`\pi`	Greek
ϖ	Variant pi	`\varpi`	Greek
ρ	Rho	`\rho`	Greek
ϱ	Variant rho	`\varrho`	Greek
σ	Sigma	`\sigma`	Greek
ς	Final sigma	`\varsigma`	Greek

Notation as shown	Called	LaTeX inserted	Menu
τ	Tau	`\tau`	Greek
υ	Upsilon	`\upsilon`	Greek
φ	Phi	`\phi`	Greek
ϕ	Variant phi	`\varphi`	Greek
χ	Chi	`\chi`	Greek
ψ	Psi	`\psi`	Greek
ω	Omega	`\omega`	Greek
ϝ	Digamma	`\digamma`	Greek
Γ	Gamma (upper)	`\Gamma`	Greek
Δ	Delta (upper)	`\Delta`	Greek
Θ	Theta (upper)	`\Theta`	Greek
Λ	Lambda (upper)	`\Lambda`	Greek
Ξ	Xi (upper)	`\Xi`	Greek
Π	Pi (upper)	`\Pi`	Greek
Σ	Sigma (upper)	`\Sigma`	Greek
ϒ	Upsilon (upper)	`\Upsilon`	Greek
Φ	Phi (upper)	`\Phi`	Greek
Ψ	Psi (upper)	`\Psi`	Greek
Ω	Omega (upper)	`\Omega`	Greek
ℵ	Aleph	`\aleph`	Greek
ℶ	Beth	`\beth`	Greek
ℷ	Gimel	`\gimel`	Greek
ℸ	Daleth	`\daleth`	Greek
ı	Dotless i	`\imath`	Greek
ȷ	Dotless j	`\jmath`	Greek
∇	Nabla	`\nabla`	Greek
∂	Partial derivative	`\partial`	Greek
ℑ	Imaginary part	`\Im`	Greek
ℑ	Imaginary part	`\image`	Greek
ℜ	Real part	`\Re`	Greek
ℜ	Real part	`\real`	Greek
ℝ	Real numbers	`\Reals`	Greek
ℝ	Real numbers	`\R`	Greek
ℝ	Real numbers	`\reals`	Greek
ℕ	Natural numbers	`\N`	Greek
ℕ	Natural numbers	`\natnums`	Greek
ℤ	Integers	`\Z`	Greek

Notation as shown	Called	LaTeX inserted	Menu
$\mathbb{C}$	Complex numbers	`\cnums`	Greek
$\mathbb{C}$	Complex numbers	`\Complex`	Greek
$\Bbbk$	Blackboard bold k	`\Bbbk`	Greek
ℓ	Script ell	`\ell`	Greek
$\hbar$	h-bar	`\hbar`	Greek
$\hslash$	h-slash	`\hslash`	Greek
$\wp$	Weierstrass p	`\wp`	Greek
$\wp$	Weierstrass p	`\weierp`	Greek
$\Game$	Game	`\Game`	Greek
$\Finv$	Turned F	`\Finv`	Greek
ð	Eth	`\eth`	Greek
Œ	OE ligature	`\OE`	Greek
ø	o slash	`\o`	Greek
Ø	O slash	`\O`	Greek
ß	Eszett	`\ss`	Greek
å	a ring	`\aa`	Greek
Å	A ring	`\AA`	Greek
ı	Dotless i	`\i`	Greek
ȷ	Dotless j	`\j`	Greek
æ	ae ligature	`\ae`	Greek
Æ	AE ligature	`\AE`	Greek
œ	oe ligature	`\oe`	Greek
$\aleph$	Aleph	`\alef`	Greek
$\aleph$	Aleph	`\alefsym`	Greek
A	Alpha	`\Alpha`	Greek
B	Beta	`\Beta`	Greek
E	Epsilon	`\Epsilon`	Greek
Z	Zeta	`\Zeta`	Greek
H	Eta	`\Eta`	Greek
I	Iota	`\Iota`	Greek
K	Kappa	`\Kappa`	Greek
M	Mu	`\Mu`	Greek
N	Nu	`\Nu`	Greek
O	Omicron	`\Omicron`	Greek
P	Rho	`\Rho`	Greek
T	Tau	`\Tau`	Greek
X	Chi	`\Chi`	Greek

Notation as shown	Called	LaTeX inserted	Menu
ο	omicron	`\omicron`	Greek
ϑ	thetasym	`\thetasym`	Greek
Γ	varGamma	`\varGamma`	Greek
Δ	varDelta	`\varDelta`	Greek
Θ	varTheta	`\varTheta`	Greek
Λ	varLambda	`\varLambda`	Greek
Ξ	varXi	`\varXi`	Greek
Π	varPi	`\varPi`	Greek
Σ	varSigma	`\varSigma`	Greek
Υ	varUpsilon	`\varUpsilon`	Greek
Φ	varPhi	`\varPhi`	Greek
Ψ	varPsi	`\varPsi`	Greek
Ω	varOmega	`\varOmega`	Greek
tilde	Tilde	`\tilde{ }`	Accents
widetilde	Wide tilde	`\widetilde{ }`	Accents
acute	Acute accent	`\acute{ }`	Accents
bar	Bar / macron	`\bar{ }`	Accents
breve	Breve	`\breve{ }`	Accents
check	Check / caron	`\check{ }`	Accents
dot	Dot above	`\dot{ }`	Accents
ddot	Double dot / umlaut	`\ddot{ }`	Accents
dddot	Triple dot	`\dddot{ }`	Accents
ddddot	Quadruple dot	`\ddddot{ }`	Accents
grave	Grave accent	`\grave{ }`	Accents
hat	Hat / circumflex	`\hat{ }`	Accents
widehat	Wide hat	`\widehat{ }`	Accents
mathring	Ring above	`\mathring{ }`	Accents
vec	Vector arrow	`\vec{ }`	Accents
overline	Overline	`\overline{ }`	Accents
underline	Underline	`\underline{ }`	Accents
underbar	Underbar	`\underbar{ }`	Accents
overbrace	Overbrace with label	`\overbrace{ }^{ }`	Accents
underbrace	Underbrace with label	`\underbrace{ }_{ }`	Accents
overbracket	Overbracket with label	`\overbracket{ }^{ }`	Accents
overbrace	Overbrace without label	`\overbrace{ }`	Accents
underbrace	Underbrace without label	`\underbrace{ }`	Accents
overbracket	Overbracket without label	`\overbracket{ }`	Accents

Notation as shown	Called	LaTeX inserted	Menu
overleftarrow	Left arrow over	`\overleftarrow{ }`	Accents
overrightarrow	Right arrow over	`\overrightarrow{ }`	Accents
underleftarrow	Left arrow under	`\underleftarrow{ }`	Accents
underrightarrow	Right arrow under	`\underrightarrow{ }`	Accents
overleftrightarrow	Bidirectional arrow over	`\overleftrightarrow{ }`	Accents
underleftrightarrow	Bidirectional arrow under	`\underleftrightarrow{ }`	Accents
overleftharpoon	Left harpoon over	`\overleftharpoon{ }`	Accents
overrightharpoon	Right harpoon over	`\overrightharpoon{ }`	Accents
acute'	Acute accent	`\'{ }`	Accents
grave`	Grave accent	`\ `{ }`	Accents
hat^	Circumflex / hat	`\^{ }`	Accents
tilde~	Tilde	`\~{ }`	Accents
macron=	Macron / bar	`\={ }`	Accents
breveu	Breve	`\u{ }`	Accents
dot.	Dot above	`\.{ }`	Accents
ringr	Ring above	`\r{ }`	Accents
doubleacuteH	Double acute	`\H{ }`	Accents
caronv	Caron / check	`\v{ }`	Accents
umlaut"	Umlaut / double dot	`\"{ }`	Accents
cancel □	cancel	`\cancel{□}`	Accents
bcancel □	bcancel	`\bcancel{□}`	Accents
xcancel □	xcancel	`\xcancel{□}`	Accents
sout □	sout	`\sout{□}`	Accents
phase □	phase	`\phase{□}`	Accents
angln □	angln	`\angln{□}`	Accents
\Bigl({#?}\Bigr)	Parentheses	`\Bigl({□}\Bigr)`	Delimiters
\Bigl[{#?}\Bigr]	Brackets	`\Bigl[{□}\Bigr]`	Delimiters
\Bigl\lbrace{#?}\Bigr\rbrace	Braces	`\Bigl\lbrace{□}\Bigr\rbrace`	Delimiters
\Bigl\langle{#?}\Bigr\rangle	Angles	`\Bigl\langle{□}\Bigr\rangle`	Delimiters
\Bigl\lvert{#?}\Bigr\rvert	Bars	`\Bigl\lvert{□}\Bigr\rvert`	Delimiters
\Bigl\lVert{#?}\Bigr\rVert	Double bars	`\Bigl\lVert{□}\Bigr\rVert`	Delimiters
\Bigl\lceil{#?}\Bigr\rceil	Ceiling	`\Bigl\lceil{□}\Bigr\rceil`	Delimiters
\Bigl\lfloor{#?}\Bigr\rfloor	Floor	`\Bigl\lfloor{□}\Bigr\rfloor`	Delimiters

Notation as shown	Called	LaTeX inserted	Menu
\Bigl\llbracket{#?}\Bigr\rrbracket	Double brackets	`\Bigl\llbracket{□}\Bigr\rrbracket`	Delimiters
Small	Delimiter size: Small	`\bigl ... \bigr`	Delimiters
Medium	Delimiter size: Medium	`\Bigl ... \Bigr`	Delimiters
Large	Delimiter size: Large	`\biggl ... \biggr`	Delimiters
Largest	Delimiter size: Largest	`\Biggl ... \Biggr`	Delimiters
\	Backslash	`\backslash`	Delimiters
∑	sum	`\sum`	Operators
∏	prod	`\prod`	Operators
∐	coprod	`\coprod`	Operators
∫	int	`\int`	Operators
∫	intop	`\intop`	Operators
∫	smallint	`\smallint`	Operators
∬	iint	`\iint`	Operators
∭	iiint	`\iiint`	Operators
∮	oint	`\oint`	Operators
∯	oiint	`\oiint`	Operators
∰	oiiint	`\oiiint`	Operators
⊗	bigotimes	`\bigotimes`	Operators
⊕	bigoplus	`\bigoplus`	Operators
⊙	bigodot	`\bigodot`	Operators
⊎	biguplus	`\biguplus`	Operators
⊔	bigsqcup	`\bigsqcup`	Operators
⋁	bigvee	`\bigvee`	Operators
⋀	bigwedge	`\bigwedge`	Operators
⋂	bigcap	`\bigcap`	Operators
⋃	bigcup	`\bigcup`	Operators
+	Plus (+)	`+`	Operators
·	cdot	`\cdot`	Operators
⋗	gtrdot	`\gtrdot`	Operators
{#?} \pmod{#?}	pmod with slots	`{□} \pmod{□}`	Operators
-	Minus (-)	`-`	Operators
·	cdotp	`\cdotp`	Operators
⊺	intercal	`\intercal`	Operators
{#?} \pod{#?}	pod with slots	`{□} \pod{□}`	Operators
/	Slash (/)	`/`	Operators
·	centerdot	`\centerdot`	Operators
∧	land	`\land`	Operators

Notation as shown	Called	LaTeX inserted	Menu
⊳	rhd	`\rhd`	Operators
*	Asterisk (*)	`*`	Operators
∘	circ	`\circ`	Operators
⋋	leftthreetimes	`\leftthreetimes`	Operators
⋌	rightthreetimes	`\rightthreetimes`	Operators
⨿	amalg	`\amalg`	Operators
⊛	circledast	`\circledast`	Operators
..	ldotp	`\ldotp`	Operators
⋊	rtimes	`\rtimes`	Operators
&	And	`\And`	Operators
⊚	circledcirc	`\circledcirc`	Operators
∨	lor	`\lor`	Operators
∖	setminus	`\setminus`	Operators
∗	ast	`\ast`	Operators
⊝	circleddash	`\circleddash`	Operators
⋖	lessdot	`\lessdot`	Operators
∖	smallsetminus	`\smallsetminus`	Operators
⊼	barwedge	`\barwedge`	Operators
⋓	Cup	`\Cup`	Operators
⊲	lhd	`\lhd`	Operators
⊓	sqcap	`\sqcap`	Operators
◯	bigcirc	`\bigcirc`	Operators
∪	cup	`\cup`	Operators
⋉	ltimes	`\ltimes`	Operators
⊔	sqcup	`\sqcup`	Operators
%	bmod	`\bmod`	Operators
⋎	curlyvee	`\curlyvee`	Operators
{#?}\mod {#?}	mod with slots	`{□}\mod {□}`	Operators
×	times	`\times`	Operators
⊡	boxdot	`\boxdot`	Operators
⋏	curlywedge	`\curlywedge`	Operators
∓	mp	`\mp`	Operators
⊴	unlhd	`\unlhd`	Operators
⊟	boxminus	`\boxminus`	Operators
÷	div	`\div`	Operators
⊙	odot	`\odot`	Operators
⊵	unrhd	`\unrhd`	Operators

Notation as shown	Called	LaTeX inserted	Menu
⊞	boxplus	`\boxplus`	Operators
⋇	divideontimes	`\divideontimes`	Operators
⊖	ominus	`\ominus`	Operators
⊎	uplus	`\uplus`	Operators
⊠	boxtimes	`\boxtimes`	Operators
∔	dotplus	`\dotplus`	Operators
⊕	oplus	`\oplus`	Operators
∨	vee	`\vee`	Operators
•	bullet	`\bullet`	Operators
⩞	doublebarwedge	`\doublebarwedge`	Operators
⊗	otimes	`\otimes`	Operators
⊻	veebar	`\veebar`	Operators
⋒	Cap	`\Cap`	Operators
⋒	doublecap	`\doublecap`	Operators
⊘	oslash	`\oslash`	Operators
∧	wedge	`\wedge`	Operators
∩	cap	`\cap`	Operators
⋓	doublecup	`\doublecup`	Operators
±	pm	`\pm`	Operators
±	plusmn	`\plusmn`	Operators
≀	wr	`\wr`	Operators
□/□	frac	`\frac{□}{□}`	Fractions
□/□	tfrac	`\tfrac{□}{□}`	Fractions
□/□	dfrac	`\dfrac{□}{□}`	Fractions
(□ over □)	binom	`\binom{□}{□}`	Fractions
=	Equals	`=`	Relations
≠	Not equal	`\ne`	Relations
≤	Less or equal	`\le`	Relations
≥	Greater or equal	`\ge`	Relations
≈	Approximately	`\approx`	Relations
∝	Proportional to	`\propto`	Relations
<	<	`<`	Relations
>	>	`>`	Relations
:	:	`:`	Relations
≤	leq	`\leq`	Relations
≥	geq	`\geq`	Relations
≠	neq	`\neq`	Relations

Notation as shown	Called	LaTeX inserted	Menu
<	lt	`\lt`	Relations
>	gt	`\gt`	Relations
≑	doteqdot	`\doteqdot`	Relations
≖	eqcirc	`\eqcirc`	Relations
=:	eqcolon	`\eqcolon`	Relations
−:	minuscolon	`\minuscolon`	Relations
=:	Eqcolon	`\Eqcolon`	Relations
∹	minuscoloncolon	`\minuscoloncolon`	Relations
=:	eqqcolon	`\eqqcolon`	Relations
=:	equalscolon	`\equalscolon`	Relations
::=	Eqqcolon	`\Eqqcolon`	Relations
::=	equalscoloncolon	`\equalscoloncolon`	Relations
≈:	approxcolon	`\approxcolon`	Relations
≈::	approxcoloncolon	`\approxcoloncolon`	Relations
≊	approxeq	`\approxeq`	Relations
≍	asymp	`\asymp`	Relations
∍	backepsilon	`\backepsilon`	Relations
∽	backsim	`\backsim`	Relations
⋍	backsimeq	`\backsimeq`	Relations
≬	between	`\between`	Relations
⋈	bowtie	`\bowtie`	Relations
≏	bumpeq	`\bumpeq`	Relations
≎	Bumpeq	`\Bumpeq`	Relations
≗	circeq	`\circeq`	Relations
:≈	colonapprox	`\colonapprox`	Relations
::≈	Colonapprox	`\Colonapprox`	Relations
::≈	coloncolonapprox	`\coloncolonapprox`	Relations
:=	coloneq	`\coloneq`	Relations
:-	colonminus	`\colonminus`	Relations
::=	Coloneq	`\Coloneq`	Relations
::-	coloncolonminus	`\coloncolonminus`	Relations
:=	coloneqq	`\coloneqq`	Relations
:=	colonequals	`\colonequals`	Relations
::=	Coloneqq	`\Coloneqq`	Relations
::=	coloncolonequals	`\coloncolonequals`	Relations
:∼	colonsim	`\colonsim`	Relations
::∼	Colonsim	`\Colonsim`	Relations

Notation as shown	Called	LaTeX inserted	Menu
∷∼	coloncolonsim	`\coloncolonsim`	Relations
≅	cong	`\cong`	Relations
⋞	curlyeqprec	`\curlyeqprec`	Relations
⋟	curlyeqsucc	`\curlyeqsucc`	Relations
⊣	dashv	`\dashv`	Relations
∷	dblcolon	`\dblcolon`	Relations
∷	coloncolon	`\coloncolon`	Relations
≐	doteq	`\doteq`	Relations
≑	Doteq	`\Doteq`	Relations
≂	eqsim	`\eqsim`	Relations
⪖	eqslantgtr	`\eqslantgtr`	Relations
⪕	eqslantless	`\eqslantless`	Relations
≡	equiv	`\equiv`	Relations
≒	fallingdotseq	`\fallingdotseq`	Relations
⌢	frown	`\frown`	Relations
≧	geqq	`\geqq`	Relations
⩾	geqslant	`\geqslant`	Relations
≫	gg	`\gg`	Relations
⋙	ggg	`\ggg`	Relations
⋙	gggtr	`\gggtr`	Relations
⪆	gtrapprox	`\gtrapprox`	Relations
⋛	gtreqless	`\gtreqless`	Relations
⪌	gtreqqless	`\gtreqqless`	Relations
≷	gtrless	`\gtrless`	Relations
≳	gtrsim	`\gtrsim`	Relations
⊷	imageof	`\imageof`	Relations
⨝	Join	`\Join`	Relations
≦	leqq	`\leqq`	Relations
⩽	leqslant	`\leqslant`	Relations
⪅	lessapprox	`\lessapprox`	Relations
⋚	lesseqgtr	`\lesseqgtr`	Relations
⪋	lesseqqgtr	`\lesseqqgtr`	Relations
≶	lessgtr	`\lessgtr`	Relations
≲	lesssim	`\lesssim`	Relations
≪	ll	`\ll`	Relations
⋘	lll	`\lll`	Relations
⋘	llless	`\llless`	Relations

Notation as shown	Called	LaTeX inserted	Menu
$\models$	models	`\models`	Relations
$\multimap$	multimap	`\multimap`	Relations
⊶	origof	`\origof`	Relations
$\owns$	owns	`\owns`	Relations
$\parallel$	parallel	`\parallel`	Relations
$\perp$	perp	`\perp`	Relations
$\pitchfork$	pitchfork	`\pitchfork`	Relations
$\prec$	prec	`\prec`	Relations
$\precapprox$	precapprox	`\precapprox`	Relations
$\preccurlyeq$	preccurlyeq	`\preccurlyeq`	Relations
$\preceq$	preceq	`\preceq`	Relations
$\precsim$	precsim	`\precsim`	Relations
$\risingdotseq$	risingdotseq	`\risingdotseq`	Relations
$\shortmid$	shortmid	`\shortmid`	Relations
$\shortparallel$	shortparallel	`\shortparallel`	Relations
$\sim$	sim	`\sim`	Relations
$\simeq$	simeq	`\simeq`	Relations
$\smallfrown$	smallfrown	`\smallfrown`	Relations
$\smallsmile$	smallsmile	`\smallsmile`	Relations
$\smile$	smile	`\smile`	Relations
$\sqsubset$	sqsubset	`\sqsubset`	Relations
$\sqsubseteq$	sqsubseteq	`\sqsubseteq`	Relations
$\sqsupset$	sqsupset	`\sqsupset`	Relations
$\sqsupseteq$	sqsupseteq	`\sqsupseteq`	Relations
$\subset$	sub	`\sub`	Relations
$\subseteq$	sube	`\sube`	Relations
$\Subset$	Subset	`\Subset`	Relations
$\subseteqq$	subseteqq	`\subseteqq`	Relations
$\succ$	succ	`\succ`	Relations
$\succapprox$	succapprox	`\succapprox`	Relations
$\succcurlyeq$	succcurlyeq	`\succcurlyeq`	Relations
$\succeq$	succeq	`\succeq`	Relations
$\succsim$	succsim	`\succsim`	Relations
$\supseteq$	supe	`\supe`	Relations
$\Supset$	Supset	`\Supset`	Relations
$\supseteqq$	supseteqq	`\supseteqq`	Relations
$\thickapprox$	thickapprox	`\thickapprox`	Relations

Notation as shown	Called	LaTeX inserted	Menu
∼	thicksim	`\thicksim`	Relations
⊴	trianglelefteq	`\trianglelefteq`	Relations
≜	triangleq	`\triangleq`	Relations
⊵	trianglerighteq	`\trianglerighteq`	Relations
∝	varpropto	`\varpropto`	Relations
△	vartriangle	`\vartriangle`	Relations
⊲	vartriangleleft	`\vartriangleleft`	Relations
⊳	vartriangleright	`\vartriangleright`	Relations
:	vcentcolon	`\vcentcolon`	Relations
:	ratio	`\ratio`	Relations
⊢	vdash	`\vdash`	Relations
⊨	vDash	`\vDash`	Relations
⊩	Vdash	`\Vdash`	Relations
⊪	Vvdash	`\Vvdash`	Relations
≠	not	`\not`	Relations
⪊	gnapprox	`\gnapprox`	Relations
⪈	gneq	`\gneq`	Relations
≩	gneqq	`\gneqq`	Relations
⋧	gnsim	`\gnsim`	Relations
≩	gvertneqq	`\gvertneqq`	Relations
⪉	lnapprox	`\lnapprox`	Relations
⪇	lneq	`\lneq`	Relations
≨	lneqq	`\lneqq`	Relations
⋦	lnsim	`\lnsim`	Relations
≨	lvertneqq	`\lvertneqq`	Relations
≇	ncong	`\ncong`	Relations
≠	ne	`\ne`	Relations
≠	neq	`\neq`	Relations
≥/	ngeq	`\ngeq`	Relations
≥/	ngeqq	`\ngeqq`	Relations
≥/	ngeqslant	`\ngeqslant`	Relations
>/	ngtr	`\ngtr`	Relations
≤/	nleq	`\nleq`	Relations
≤/	nleqq	`\nleqq`	Relations
≤/	nleqslant	`\nleqslant`	Relations
</	nless	`\nless`	Relations
∤	nmid	`\nmid`	Relations

Notation as shown	Called	LaTeX inserted	Menu
∉	notin	`\notin`	Relations
∌	notni	`\notni`	Relations
∦	nparallel	`\nparallel`	Relations
⊀	nprec	`\nprec`	Relations
⋠	npreceq	`\npreceq`	Relations
∤	nshortmid	`\nshortmid`	Relations
∦	nshortparallel	`\nshortparallel`	Relations
≁	nsim	`\nsim`	Relations
⊈	nsubseteq	`\nsubseteq`	Relations
⊈	nsubseteqq	`\nsubseteqq`	Relations
⊁	nsucc	`\nsucc`	Relations
⋡	nsucceq	`\nsucceq`	Relations
⊉	nsupseteq	`\nsupseteq`	Relations
⊉	nsupseteqq	`\nsupseteqq`	Relations
⋪	ntriangleleft	`\ntriangleleft`	Relations
⋬	ntrianglelefteq	`\ntrianglelefteq`	Relations
⋫	ntriangleright	`\ntriangleright`	Relations
⋭	ntrianglerighteq	`\ntrianglerighteq`	Relations
⊬	nvdash	`\nvdash`	Relations
⊭	nvDash	`\nvDash`	Relations
⊯	nVDash	`\nVDash`	Relations
⊮	nVdash	`\nVdash`	Relations
⪹	precnapprox	`\precnapprox`	Relations
⪵	precneqq	`\precneqq`	Relations
⋨	precnsim	`\precnsim`	Relations
⊊	subsetneq	`\subsetneq`	Relations
⫋	subsetneqq	`\subsetneqq`	Relations
⪺	succnapprox	`\succnapprox`	Relations
⪶	succneqq	`\succneqq`	Relations
⋩	succnsim	`\succnsim`	Relations
⊋	supsetneq	`\supsetneq`	Relations
⫌	supsetneqq	`\supsetneqq`	Relations
⊊	varsubsetneq	`\varsubsetneq`	Relations
⫋	varsubsetneqq	`\varsubsetneqq`	Relations
⊋	varsupsetneq	`\varsupsetneq`	Relations
⫌	varsupsetneqq	`\varsupsetneqq`	Relations
∈	Element of	`\in`	Sets and Logic

Notation as shown	Called	LaTeX inserted	Menu
$\notin$	Not element of	`\notin`	Sets and Logic
$\emptyset$	Empty set	`\emptyset`	Sets and Logic
$\subset$	Subset	`\subset`	Sets and Logic
$\subseteq$	Subset or equal	`\subseteq`	Sets and Logic
$\supset$	Superset	`\supset`	Sets and Logic
$\cup$	Union	`\cup`	Sets and Logic
$\cap$	Intersection	`\cap`	Sets and Logic
$\forall$	For all	`\forall`	Sets and Logic
$\exists$	There exists	`\exists`	Sets and Logic
$\neg$	Logical not	`\neg`	Sets and Logic
$\wedge$	Logical and	`\wedge`	Sets and Logic
$\vee$	Logical or	`\vee`	Sets and Logic
$\rightarrow$	Right arrow	`\rightarrow`	Arrows
$\leftarrow$	Left arrow	`\leftarrow`	Arrows
$\leftrightarrow$	Both arrows	`\leftrightarrow`	Arrows
$\uparrow$	Up arrow	`\uparrow`	Arrows
$\downarrow$	Down arrow	`\downarrow`	Arrows
$\updownarrow$	Up-down arrow	`\updownarrow`	Arrows
$\Rightarrow$	Implies	`\Rightarrow`	Arrows
$\Leftarrow$	Implied by	`\Leftarrow`	Arrows
$\Leftrightarrow$	If and only if	`\Leftrightarrow`	Arrows
$\Uparrow$	Double up arrow	`\Uparrow`	Arrows
$\Downarrow$	Double down arrow	`\Downarrow`	Arrows
$\Updownarrow$	Double up-down arrow	`\Updownarrow`	Arrows
$\mapsto$	Maps to	`\mapsto`	Arrows
$\circlearrowleft$	circlearrowleft	`\circlearrowleft`	Arrows
$\circlearrowright$	circlearrowright	`\circlearrowright`	Arrows
$\curvearrowleft$	curvearrowleft	`\curvearrowleft`	Arrows
$\curvearrowright$	curvearrowright	`\curvearrowright`	Arrows
$\dashleftarrow$	dashleftarrow	`\dashleftarrow`	Arrows
$\dashrightarrow$	dashrightarrow	`\dashrightarrow`	Arrows
$\downdownarrows$	downdownarrows	`\downdownarrows`	Arrows
$\downharpoonleft$	downharpoonleft	`\downharpoonleft`	Arrows
$\downharpoonright$	downharpoonright	`\downharpoonright`	Arrows
$\gets$	gets	`\gets`	Arrows
$\hookleftarrow$	hookleftarrow	`\hookleftarrow`	Arrows
$\hookrightarrow$	hookrightarrow	`\hookrightarrow`	Arrows

Notation as shown	Called	LaTeX inserted	Menu
$\iff$	iff	`\iff`	Arrows
$\impliedby$	impliedby	`\impliedby`	Arrows
$\implies$	implies	`\implies`	Arrows
$\leadsto$	leadsto	`\leadsto`	Arrows
$\Leftarrow$	Leftarrow	`\Leftarrow`	Arrows
$\leftarrowtail$	leftarrowtail	`\leftarrowtail`	Arrows
$\leftharpoondown$	leftharpoondown	`\leftharpoondown`	Arrows
$\leftharpoonup$	leftharpoonup	`\leftharpoonup`	Arrows
$\leftleftarrows$	leftleftarrows	`\leftleftarrows`	Arrows
$\leftrightarrows$	leftrightarrows	`\leftrightarrows`	Arrows
$\leftrightharpoons$	leftrightharpoons	`\leftrightharpoons`	Arrows
$\leftrightsquigarrow$	leftrightsquigarrow	`\leftrightsquigarrow`	Arrows
$\Lleftarrow$	Lleftarrow	`\Lleftarrow`	Arrows
$\longleftarrow$	longleftarrow	`\longleftarrow`	Arrows
$\Longleftarrow$	Longleftarrow	`\Longleftarrow`	Arrows
$\longleftrightarrow$	longleftrightarrow	`\longleftrightarrow`	Arrows
$\Longleftrightarrow$	Longleftrightarrow	`\Longleftrightarrow`	Arrows
$\longmapsto$	longmapsto	`\longmapsto`	Arrows
$\longrightarrow$	longrightarrow	`\longrightarrow`	Arrows
$\Longrightarrow$	Longrightarrow	`\Longrightarrow`	Arrows
$\looparrowleft$	looparrowleft	`\looparrowleft`	Arrows
$\looparrowright$	looparrowright	`\looparrowright`	Arrows
$\Lsh$	Lsh	`\Lsh`	Arrows
$\mapsto$	mapsto	`\mapsto`	Arrows
$\nearrow$	nearrow	`\nearrow`	Arrows
$\nleftarrow$	nleftarrow	`\nleftarrow`	Arrows
$\nLeftarrow$	nLeftarrow	`\nLeftarrow`	Arrows
$\nleftrightarrow$	nleftrightarrow	`\nleftrightarrow`	Arrows
$\nLeftrightarrow$	nLeftrightarrow	`\nLeftrightarrow`	Arrows
$\nrightarrow$	nrightarrow	`\nrightarrow`	Arrows
$\nRightarrow$	nRightarrow	`\nRightarrow`	Arrows
$\nwarrow$	nwarrow	`\nwarrow`	Arrows
$\restriction$	restriction	`\restriction`	Arrows
$\rightarrowtail$	rightarrowtail	`\rightarrowtail`	Arrows
$\rightharpoondown$	rightharpoondown	`\rightharpoondown`	Arrows
$\rightharpoonup$	rightharpoonup	`\rightharpoonup`	Arrows
$\rightleftarrows$	rightleftarrows	`\rightleftarrows`	Arrows

Notation as shown	Called	LaTeX inserted	Menu
⇌	rightleftharpoons	`\rightleftharpoons`	Arrows
⇉	rightrightarrows	`\rightrightarrows`	Arrows
⇝	rightsquigarrow	`\rightsquigarrow`	Arrows
⇛	Rrightarrow	`\Rrightarrow`	Arrows
↱	Rsh	`\Rsh`	Arrows
↘	searrow	`\searrow`	Arrows
↙	swarrow	`\swarrow`	Arrows
→	to	`\to`	Arrows
↞	twoheadleftarrow	`\twoheadleftarrow`	Arrows
↠	twoheadrightarrow	`\twoheadrightarrow`	Arrows
↿	upharpoonleft	`\upharpoonleft`	Arrows
↾	upharpoonright	`\upharpoonright`	Arrows
⇈	upuparrows	`\upuparrows`	Arrows
⟨□⟩	leftlangle{#?}rightrangle	`\left\langle{□}\right\rangle`	Notation
|□⟩	leftlvert{#?}rightrangle	`\left\lvert{□}\right\rangle`	Notation
⟨□|	leftlangle{#?}rightrvert	`\left\langle{□}\right\rvert`	Notation
{□}	leftlbrace{#?}rightrbrace	`\left\lbrace{□}\right\rbrace`	Notation
∁	complement	`\complement`	Notation
∴	therefore	`\therefore`	Notation
∵	because	`\because`	Notation
∅	emptyset	`\emptyset`	Notation
∅	varnothing	`\varnothing`	Notation
∃	exists	`\exists`	Notation
∄	nexists	`\nexists`	Notation
∣	mid	`\mid`	Notation
∈	in	`\in`	Notation
∋	ni	`\ni`	Notation
¬	lnot	`\lnot`	Notation
↲	Newline	`\\`	Layout
↲	Newline	`\newline`	Layout
thin	Thin space	`\,`	Layout
thin	Thin space	`\thinspace`	Layout
med	Medium space	`\:`	Layout
med	Medium space	`\medspace`	Layout
thick	Thick space	`\;`	Layout

Notation as shown	Called	LaTeX inserted	Menu
thick	Thick space	`\thickspace`	Layout
en	En space	`\enspace`	Layout
quad	Quad space	`\quad`	Layout
qquad	Double quad space	`\qquad`	Layout
tight	Negative thin space	`\!`	Layout
tight	Negative thin space	`\negthinspace`	Layout
~	Non-breaking space (~)	`~`	Layout
boxed □	boxed	`\boxed{□}`	Layout
□ over □	stackrel	`\stackrel{□}{□}`	Layout
phantom □	phantom	`\phantom{□}`	Layout
horizontal phantom □	hphantom	`\hphantom{□}`	Layout
vertical phantom □	vphantom	`\vphantom{□}`	Layout
horizontal space □	hspace	`\hspace{□}`	Layout
horizontal space* □	hspace*	`\hspace*{□}`	Layout
arcsin	arcsin	`\arcsin`	Functions
arccos	arccos	`\arccos`	Functions
arctan	arctan	`\arctan`	Functions
arctg	arctg	`\arctg`	Functions
arcctg	arcctg	`\arcctg`	Functions
arg	arg	`\arg`	Functions
ch	ch	`\ch`	Functions
cos	cos	`\cos`	Functions
cosec	cosec	`\cosec`	Functions
cosh	cosh	`\cosh`	Functions
cot	cot	`\cot`	Functions
cotg	cotg	`\cotg`	Functions
coth	coth	`\coth`	Functions
csc	csc	`\csc`	Functions
ctg	ctg	`\ctg`	Functions
cth	cth	`\cth`	Functions
deg	deg	`\deg`	Functions
dim	dim	`\dim`	Functions
exp	exp	`\exp`	Functions
hom	hom	`\hom`	Functions
ker	ker	`\ker`	Functions
lg	lg	`\lg`	Functions
ln	ln	`\ln`	Functions

Notation as shown	Called	LaTeX inserted	Menu
log	log	`\log`	Functions
sec	sec	`\sec`	Functions
sin	sin	`\sin`	Functions
sinh	sinh	`\sinh`	Functions
sh	sh	`\sh`	Functions
tan	tan	`\tan`	Functions
tanh	tanh	`\tanh`	Functions
tg	tg	`\tg`	Functions
th	th	`\th`	Functions
f	f	`\operatorname{f}`	Functions
argmax	argmax	`\argmax`	Functions
argmin	argmin	`\argmin`	Functions
det	det	`\det`	Functions
gcd	gcd	`\gcd`	Functions
inf	inf	`\inf`	Functions
injlim	injlim	`\injlim`	Functions
lim	lim	`\lim`	Functions
liminf	liminf	`\liminf`	Functions
limsup	limsup	`\limsup`	Functions
max	max	`\max`	Functions
min	min	`\min`	Functions
plim	plim	`\plim`	Functions
Pr	Pr	`\Pr`	Functions
projlim	projlim	`\projlim`	Functions
sup	sup	`\sup`	Functions
varinjlim	varinjlim	`\varinjlim`	Functions
varliminf	varliminf	`\varliminf`	Functions
varlimsup	varlimsup	`\varlimsup`	Functions
varprojlim	varprojlim	`\varprojlim`	Functions
f*	f	`\operatorname*{f}`	Functions
f limits	f	`\operatornamewithlimits{f}`	Functions